

Biology Midterm — Student QWB_S07

Q1: Describe the structure and function of the cell membrane.

The cell membrane is a selectively permeable phospholipid bilayer responsible for regulating the transport of substances into and out of the cell, containing embedded proteins that facilitate transport and cellular signaling.

Q2: Explain the role of DNA in a cell.

Mitochondria are the powerhouse of the cell, producing energy through cellular respiration in the form of ATP.

Q3: Describe the process of photosynthesis.

Plants perform photosynthesis by using light energy to convert carbon dioxide and water into glucose and oxygen, mostly within the chloroplasts using chlorophyll.